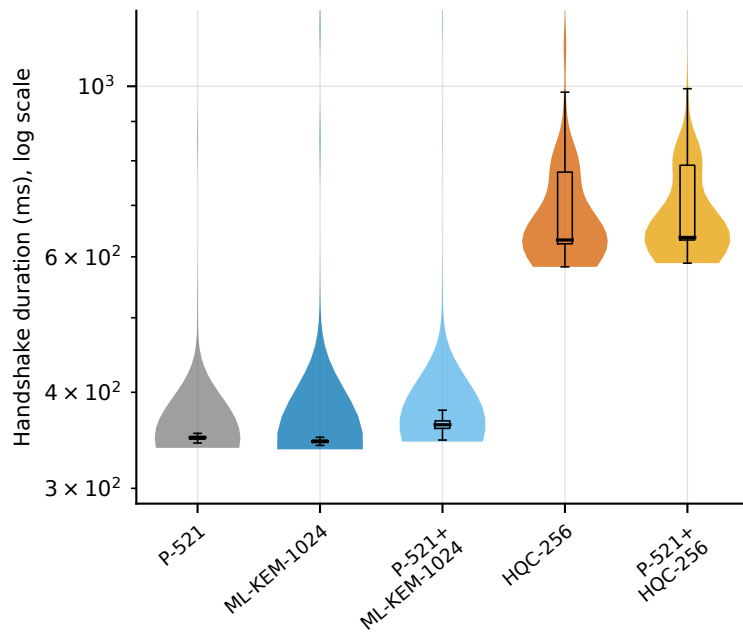
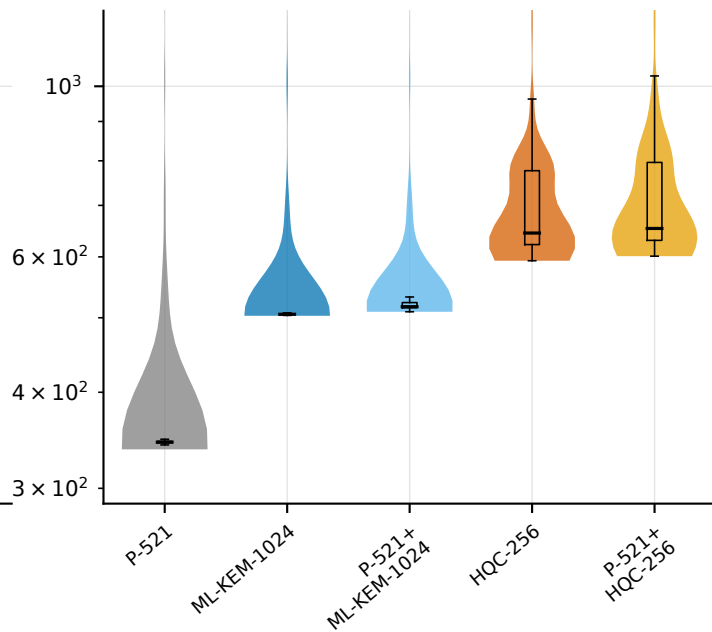


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	374.4	39.9	1.38k	400.8	56.2	2.42k
ML-KEM-1024	379.9	49.8	2.07k	549.7	36.2	1.90k
P-521+ML-KEM-1024	392.4	43.3	1.57k	560.4	34.1	2.22k
HQC-256	702.6	27.3	2.14k	708.3	26.6	2.78k
P-521+HQC-256	705.0	26.6	2.52k	734.7	29.3	2.25k